Day 1: Osmosis in Potato Strips

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Definition of Osmosis

Osmosis is defined as the net movement of water across a partially permeable membrane, proceeding from an area of high water potential to an area of low water potential.

Experimental Method

Potato strips were soaked for 30 minutes in different concentrations of sucrose solutions. The mass of the strips was measured before and after soaking to determine the percentage change in mass.

Experimental Results

- 0.0 M sucrose solution: +9 percent change in mass.
- 0.4 M sucrose solution: 0 percent change in mass (no net change).
- 0.8 M sucrose solution: -14 percent change in mass (strips were limp and plasmolysed).

Conclusions and Key Terms

Potato cells are isotonic with approximately 0.4 M sucrose solution. In 0.8 M sucrose, the external solution has a lower water potential, so cells lost water.

- Water potential: The tendency of water molecules to move from one area to another.
- Isotonic: A solution that has the same solute concentration and water potential as the cell contents, leading to no net water movement.
- Hypotonic: A solution with a lower solute concentration and higher water potential than the cell, causing net water uptake.
- Hypertonic: A solution with a higher solute concentration and lower water potential than the cell, causing net water loss.
- Turgor: The internal fluid pressure pressing the cell contents against the cell wall.
- Plasmolysis: The process where plant cells lose water in a hypertonic solution, causing cell contents to shrink away from the wall and leaving tissue limp.